

Why India Must Build Its Own **SILICON ECOSYSTEM** To Achieve Technological Control



India is building fabs, but real control lies deeper. So what will it take to truly own technology, beyond chips and into systems?



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Who really controls technology? When you buy a car today, who truly controls it? It may seem like the manufacturer or OEM does, but in reality control increasingly lies elsewhere: embedded software, proprietary hardware modules, and opaque ‘black box’ systems. These components often originate from foreign vendors, leaving users and even manufacturers without full control over functionality, bug fixes, or security. This raises a fundamental concern: are we merely users of technology, or do we truly own it? India’s growing push towards indigenous semiconductor design and system-level control stems from this very question.

The core problem

Across critical sectors such as automotive, defence, industrial automation, and the Internet of Things (IoT), India

faces a deep rooted structural challenge: systemic dependence on foreign intellectual property (IP). Much of the core technology powering these domains, ranging from microcontrollers and communication interfaces to safety modules and firmware stacks, is designed, owned, and controlled by external entities. While these imports enable rapid development and deployment, they come at the cost of limited transparency and diminished control over how systems function at their most fundamental level.

This dependence creates cascading limitations. Indian developers and system integrators often lack access to low-level architecture, firmware layers, and internal design documentation, making even minor modifications or optimisations difficult. When failures occur, whether due to bugs, performance issues, or unexpected behaviour, teams cannot independently diagnose or resolve them, and